

AVISHKAR MAHALINGPURE

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Profile

Software Engineer and MSc Computer Science student at the University of Strathclyde with strong expertise in full-stack development (Java, TypeScript, React). Proven experience building enterprise ERP systems and scalable web applications. Passionate about developer tooling and AI integration, seeking a Graduate Software Engineering role to leverage skills in backend architecture and modern frontend frameworks.

Education

University of Strathclyde

MSc in Advanced Computer Science

Glasgow, UK

Expected 2026

Sharad Institute Of Technology

Bachelor of Science in Computer Science

India

2025

Experience

Eduplus Campus (Edutantra)

Dec 2024 – June 2025

Software Engineer Intern

Pune, India

- Engineered backend modules for a large-scale ERP system tailored for universities and businesses, utilizing Groovy and Gradle for scalable build automation.
- Modernized the legacy frontend architecture, designing a sleek, accessible UI that improved user navigation and visual consistency.
- Collaborated with cross-functional teams to ensure seamless integration between backend logic and the new frontend interface.

Eduplus Campus (Edutantra)

Dec 2023 – Mar 2024

Software Engineer Intern

Pune, India

- Developed a fully functional cross-platform mobile application using Flutter, taking ownership of the full lifecycle from design to deployment.
- Implemented RESTful API integrations to fetch and display real-time database content, optimizing data flow for mobile performance.
- Delivered clean, well-documented code and conducted a comprehensive handover to the engineering team to facilitate future maintenance.

Projects

OpenBook: Local-AI Study Assistant | *FastAPI, React, TypeScript, Zustand, Ollama*

- Built an open-source, privacy-first study assistant that runs entirely on local hardware, integrating Ollama-served LLMs – including vision-capable models – instead of external AI APIs.
- Architected a normalized Zustand store using flat entity tables and id-array references, replacing a deeply nested state shape to improve scalability and reduce re-renders.
- Developed a FastAPI backend with model-capability detection, per-request Ollama URL/model overrides, and a full notes system with debounced autosave and image upload with vision-model-backed chat queries.

Medicare: AI-Powered Healthcare | *React, Node.js, Redis, Gemini API*

- Architected a secure healthcare management system featuring Role-Based Access Control (RBAC) for doctors and patients.
- Integrated Google Gemini API to build an intelligent patient summary system, allowing doctors to query medical history via Natural Language Processing (NLP).
- Optimized system performance by implementing Redis caching, significantly reducing database load and API response times.
- Engineered complex state management using Zustand to handle sensitive medical records and lab results securely.
- Integrated a secure subscription payment process using the Stripe API to handle recurring billing for premium healthcare features.

HTML Wrapper: VS Code Extension | *TypeScript, VS Code API*

- Developed and published a developer productivity tool that automates the wrapping of HTML elements, solving a common pain point for frontend engineers.
- Utilized the VS Code API to manipulate the editor selection, allowing users to seamlessly wrap code blocks with a single command.

CheckIn: Parent Student Management | *Spring Boot, PostgreSQL, React, Zustand*

- Built a full-stack family productivity platform enabling parents to assign tasks to children with deadlines, priorities, and descriptions.
- Designed a real-time progress tracking dashboard so parents can monitor task completion status and activity history per child.
- Implemented role-based access using Spring Security, separating parent and child views with distinct permissions and capabilities.
- Managed client-side state with Zustand, handling task lists, progress updates, and user sessions efficiently across the React frontend.
- Modelled relational data in PostgreSQL with parent-child account linking, task assignments, and progress logs for persistent history.

Technical Skills

Languages: Java, TypeScript, JavaScript, Groovy, C#

Backend: Spring Boot, Node.js, Express, FastAPI, Gradle

Frontend: React, Redux, Zustand, Tailwind CSS, Radix UI, ShadCN, Flutter

Databases: PostgreSQL, MongoDB, MySQL, Redis

Tools: Git, GitHub, Docker, VS Code Extensions, AI APIs (Google Gemini, Ollama), Unity